Graphical Method

General form of LPP:

$$\boldsymbol{Maximize\ or\ Minimize}\quad Z = c_1x_! + c_2x_2 + \cdots + c_nx_n$$

Subject to:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &\quad (\leq, =, \geq)\ b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &\quad (\leq, =, \geq)\ b_2 \\ &\dots \\ a_{,m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &\quad (\leq, =, \geq)\ b_m \\ x_1, x_2, \dots, x_n &\geq 0. \end{aligned}$$

Matrix form of LPP:

Maximize or Minimize $Z = C^T X$

Subject to:

$$AX\ (\leq, =, \geq)\ b,$$
$$X \geq 0.$$

Where $C^T = (c_1, c_2, \dots, c_n), X^T = (x_1, x_2, \dots, x_2), A = \left(a_{ij}\right)_{m\times n}$ **and** $b^T = (b_1, b_2, \dots, b_m)$.

Feasible Solution: A point $X^* = (x_1^*, x_2^*, \dots, x_n^*)$ satisfying all the constraint

$$AX^* \ (\leq, =, \geq) \ b,$$

together with the nonnegative condition $X^* \geq 0$, is called feasible solution.

Feasible Region: The collection of all feasible solution is called feasible set or the feasible region. It is normally denoted by S.

If S is empty, then the LPP is called infeasible.

Optimal Solution: Any feasible solution that optimize the objective function, is called optimal solution.

Unbounded Solution: If the feasible region is unbounded and the objective function is unbounded in that feasible region, then we called the LPP is unbounded.

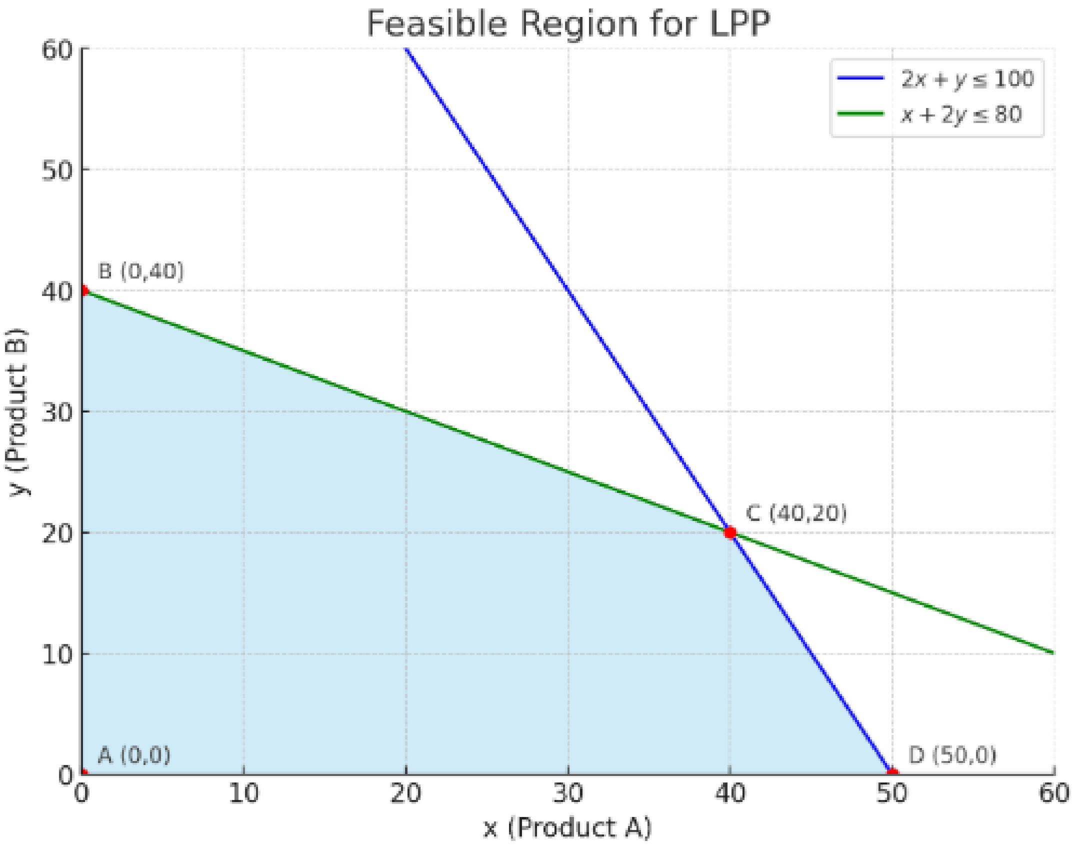
But if the objective function is bounded in that unbounded feasible region, then we say that the LPP is bounded.

Solve the following LPP using graphical method

$$\textbf{Max } Z = 40x + 50y$$

$$\textbf{Subject to, } \quad 2x + y \le 100,$$

$$x + 2y \le 80, \qquad x, y \ge 0.$$



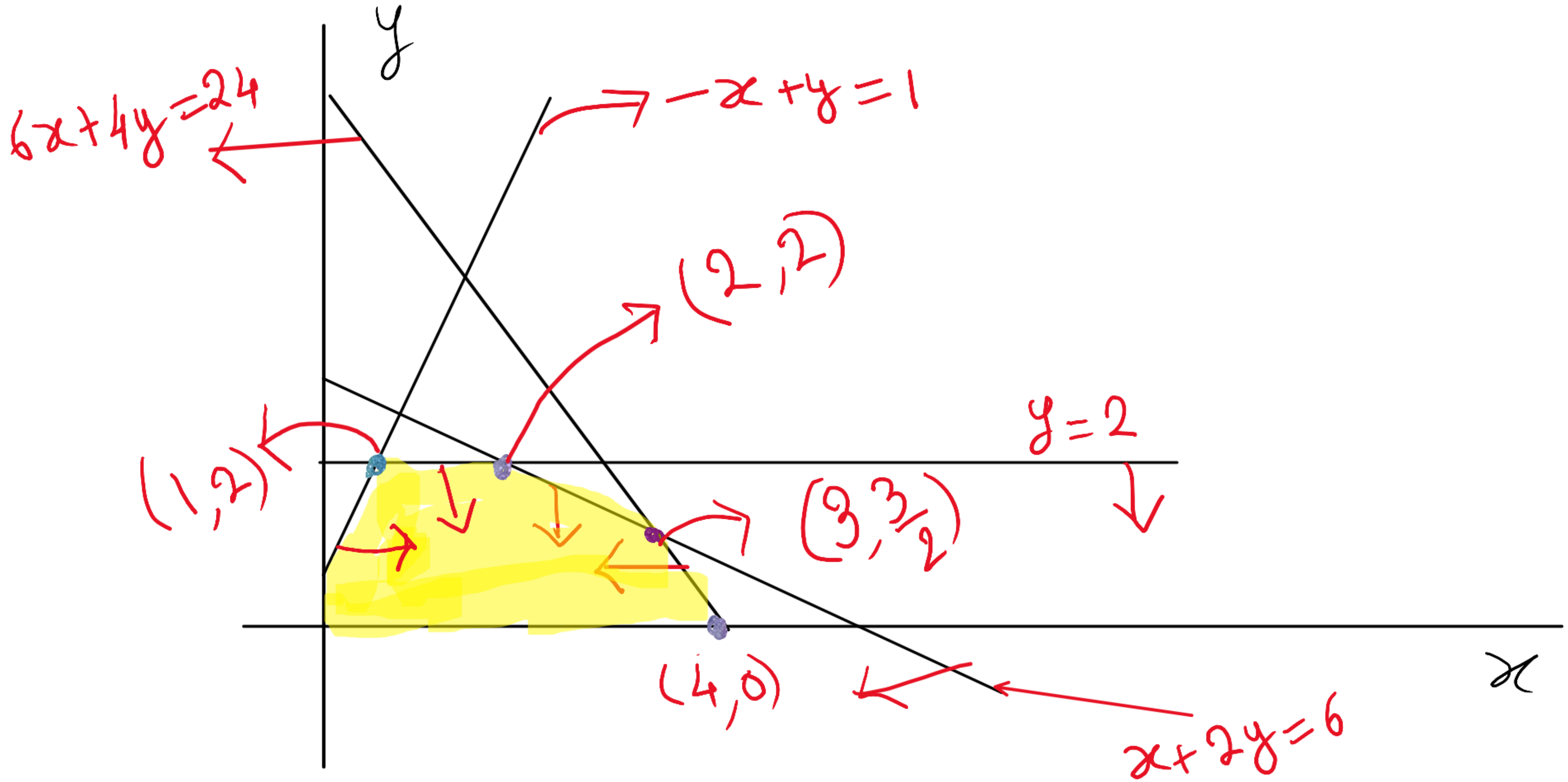
$$Z(50,0) = 2000,$$
$$Z(0,40) = 2000,$$
$$\boldsymbol{Z(40,20) = 2600}.$$

Ex 2:

$$Max\ Z = 5x + 4y$$

Subject to:

$$6x + 4y \leq 24$$
$$x + 2y \leq 6$$
$$-x + y \leq 1$$
$$y \leq 2, \quad x, y \geq 0$$



$Z(1,2) = 13$

$Z(4,0) = 20$

$Z(2,2) = 18$

$Z(3, \frac{3}{2}) = 21$

$\therefore x = 3,\ y = \frac{3}{2}$

Max $Z = 21$

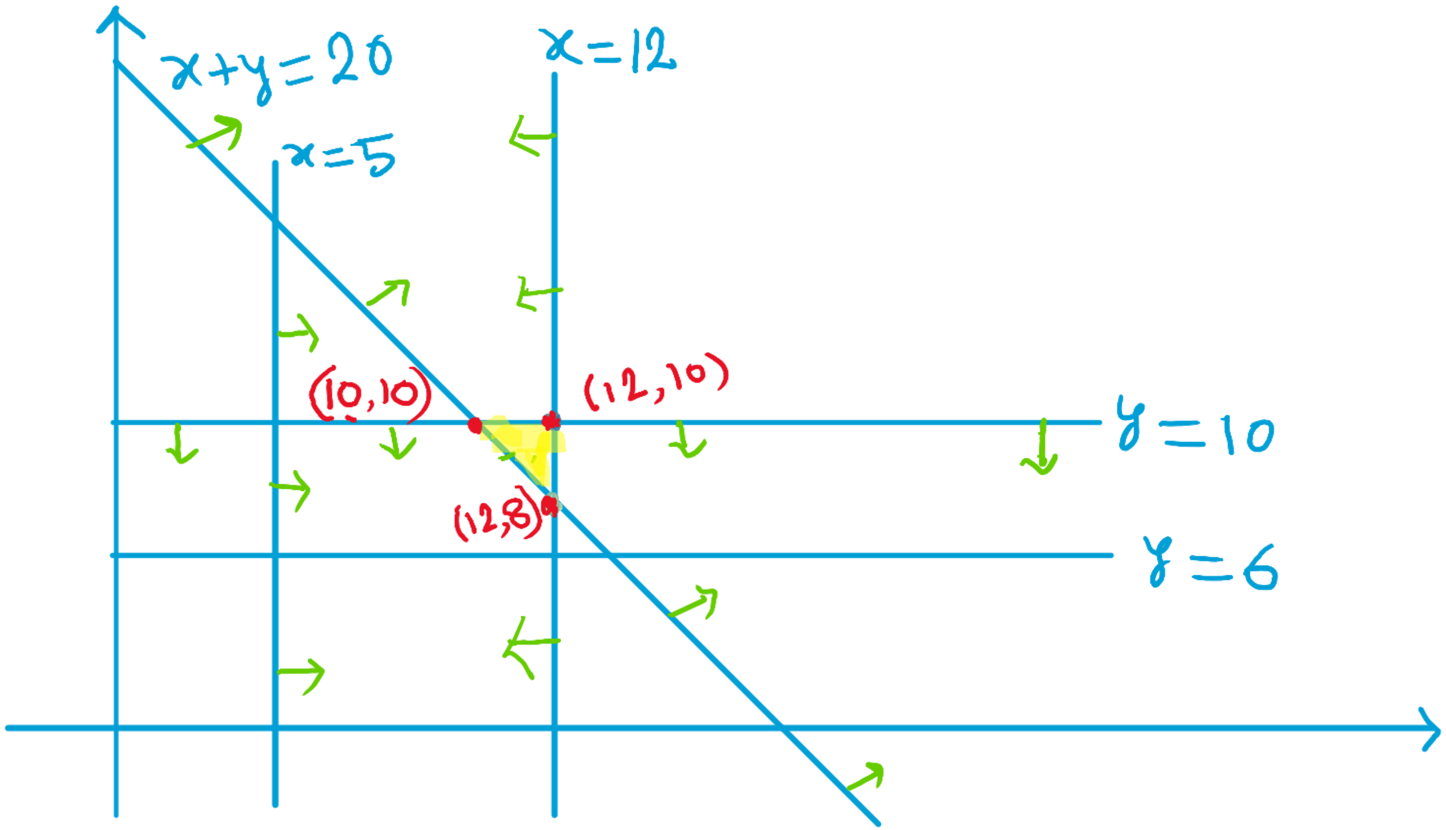
Ex: $\min Z = 8x + 6y$

Sub to: $x + y \geq 20$, $\quad y \leq 10$,

$x \leq 12$, $\quad y \geq 6$,

$x \geq 5$, $\quad x \geq 0, y \geq 0$.

Soln:



$Z(10,10) = 140$

$Z(12,10) = 156$

$Z(12,8) = 144$

$\therefore x = 10, \; y = 10$

$\min Z = 140$

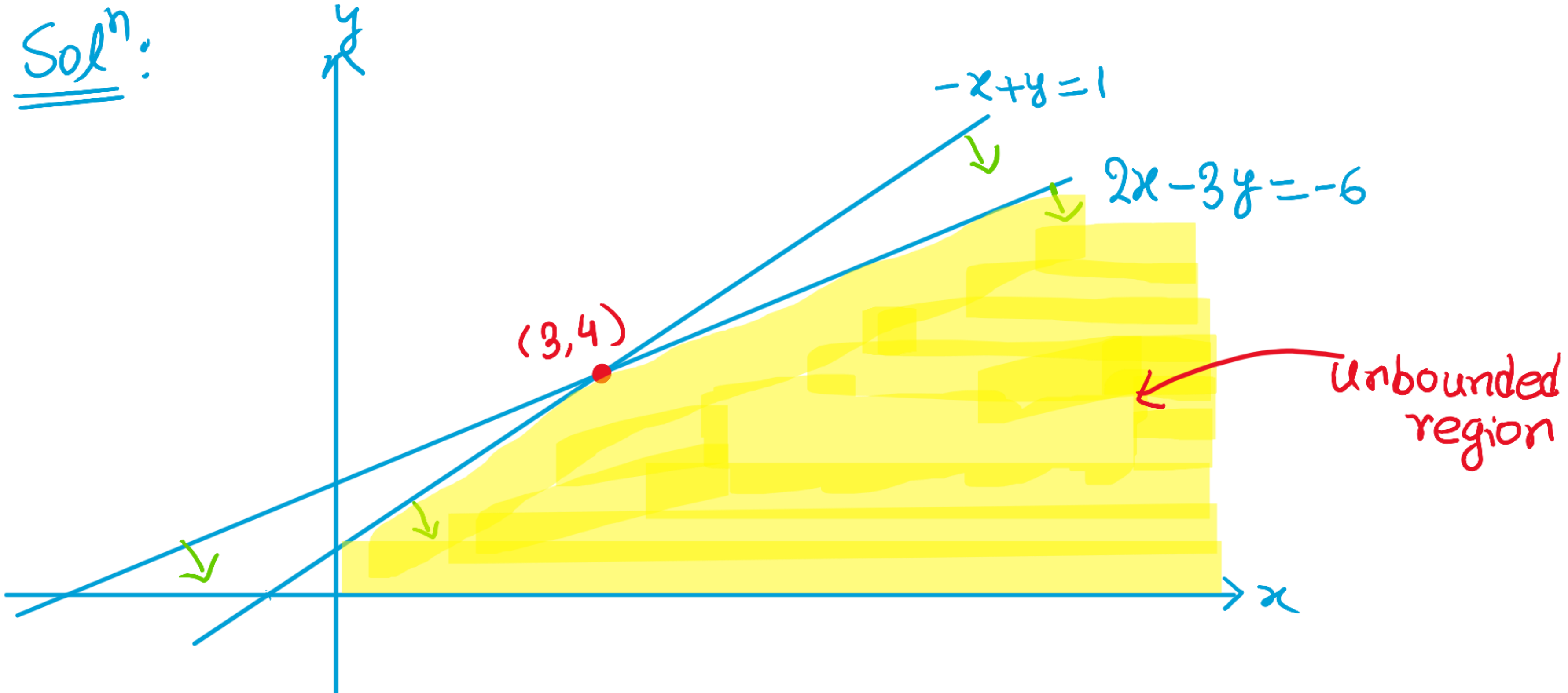
Ex:

$$\text{Min } Z = -4x + 6y$$

Sub to: $2x - 3y \geq -6$,

$-x + y \leq 1$, $x \geq 0$, $y \geq 0$.

Sol^n:



Here, $Z \to -\infty$ as $x \to \infty$

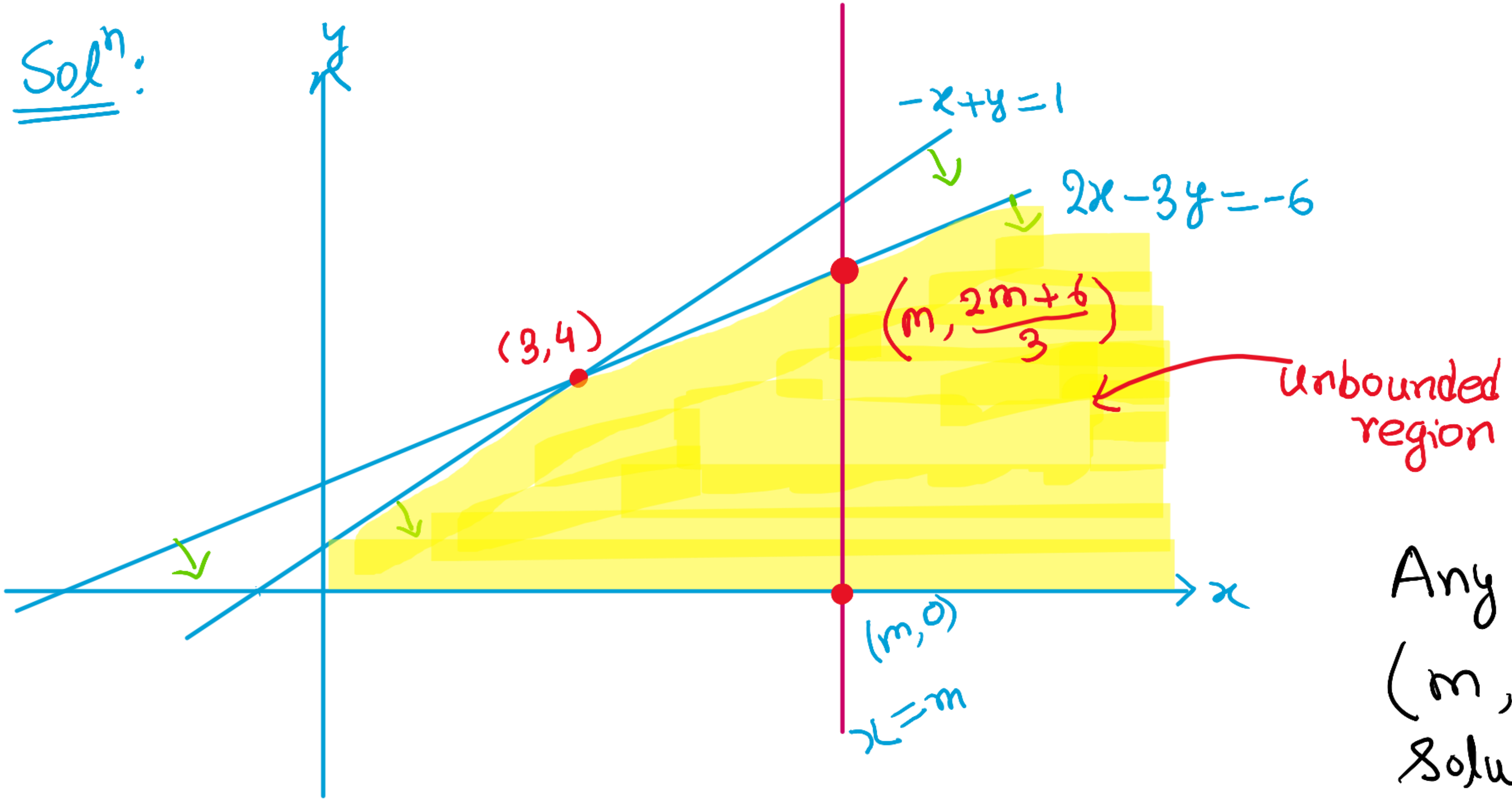
$\therefore$ The LPP is unbounded.

Ex: Max $Z = -4x + 6y$

Sub to: $2x - 3y \geq -6$,

$-x + y \leq 1$, $x \geq 0$, $y \geq 0$.

Sol^n:



$Z(m,0) = -4m$

$Z\left(m, \frac{2m+6}{3}\right) = 12$

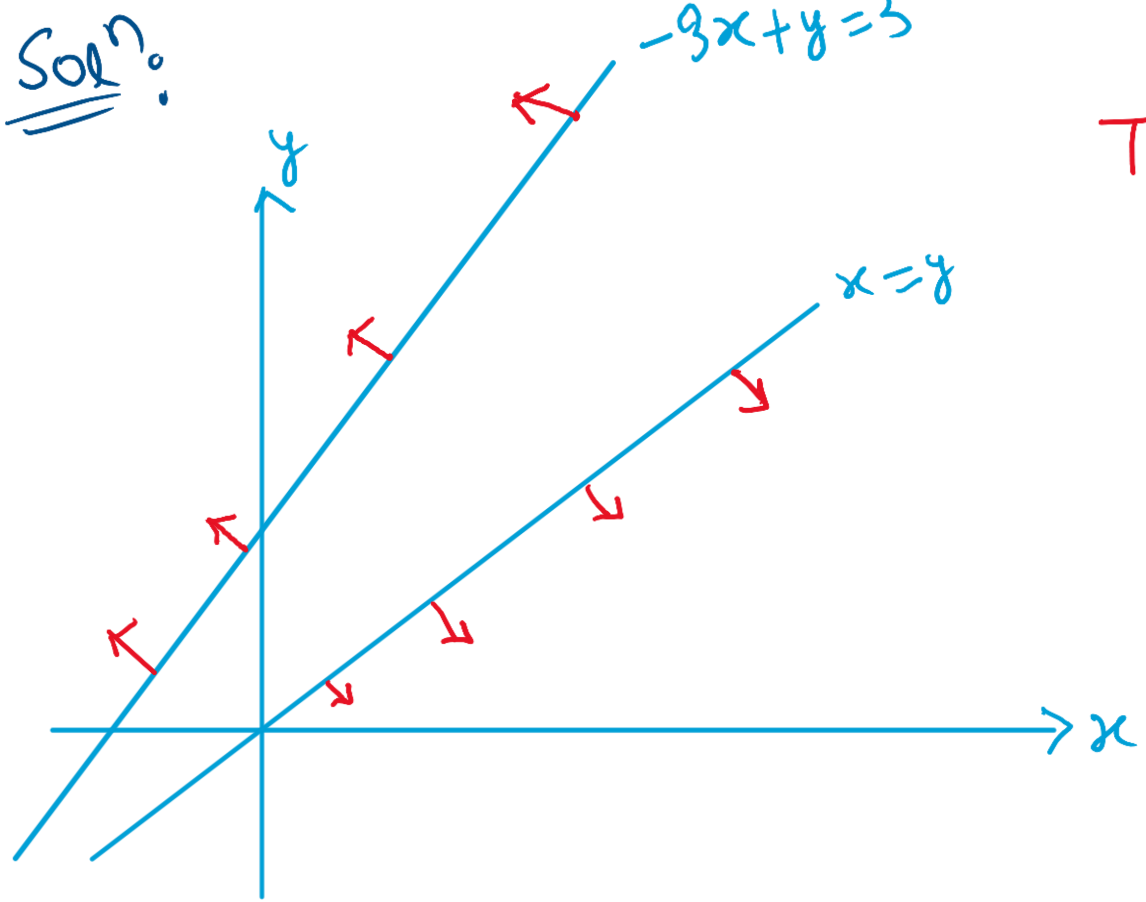
Any point of the form $\left(m, \frac{2m+6}{3}\right)$ is an optimal solution. and $Z_{max} = 12$.

Ex: Max $Z = x + y$

Sub to: $x - y \geqslant 0$,

$-3x + y \geqslant 3$, $x \geqslant 0$, $y \geqslant 0$.

Sol^n:



The constraints have no common region.

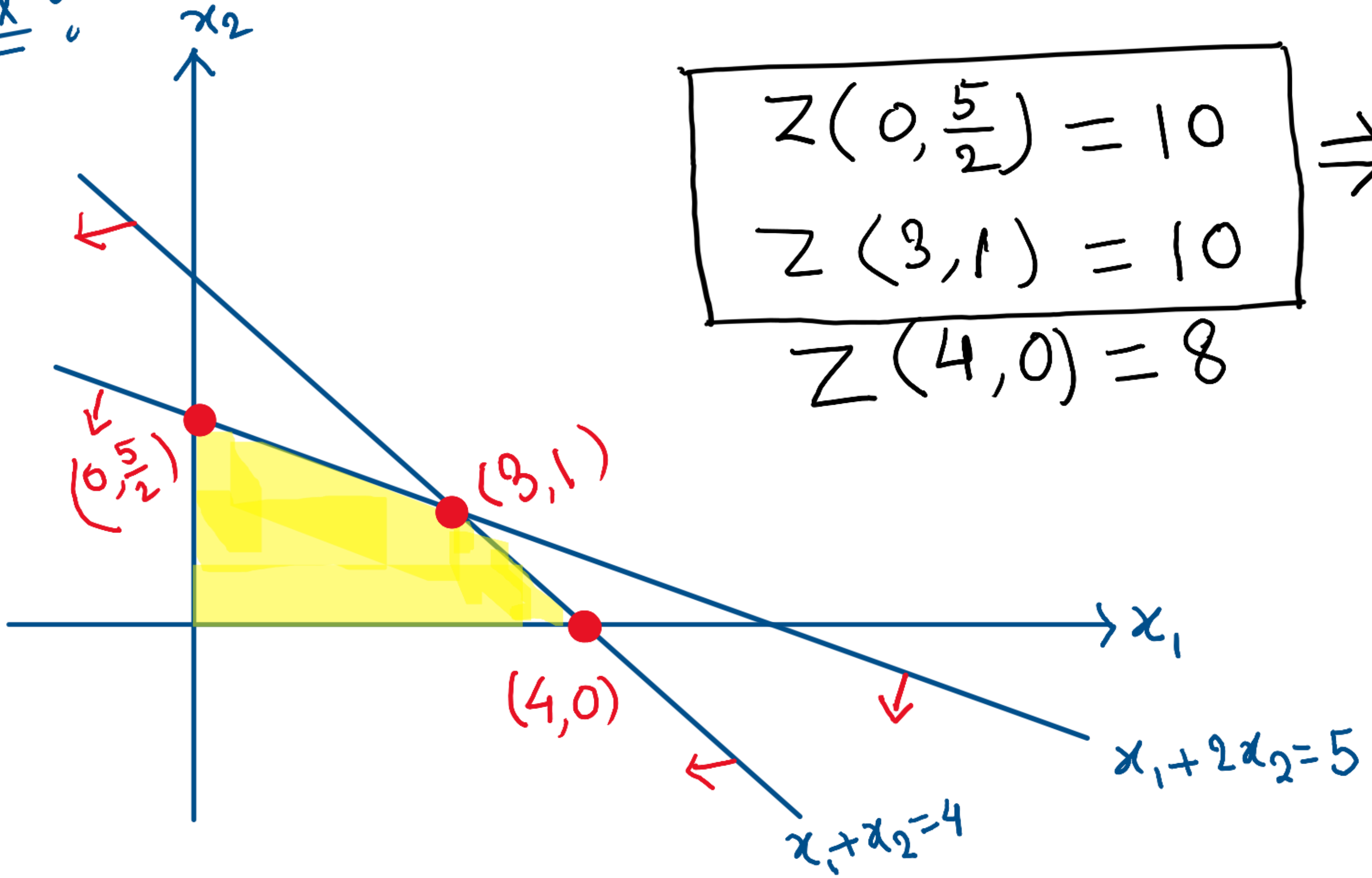
$\Rightarrow$ The region is infeasible.

$\Rightarrow$ The LPP has infeasible sol^n.

Ex: Max $Z = 2x_1 + 4x_2$

Sub to: $x_1 + 2x_2 \le 5$, $x_1 + x_2 \le 4$, $x_1 \ge 0$, $x_2 \ge 0$.

Sol^n:



$$\boxed{\begin{aligned} Z(0, \tfrac{5}{2}) &= 10 \\ Z(3,1) &= 10 \end{aligned}} \Rightarrow \text{alternative / multiple solution exist.}$$

$$Z(4,0) = 8$$

The solution of the form,

$$(x, y) = \lambda(0, \tfrac{5}{2}) + (1-\lambda)(3,1)$$

$$= \left(3 - 3\lambda, \frac{3\lambda + 2}{2}\right)$$

$$0 \le \lambda \le 1$$

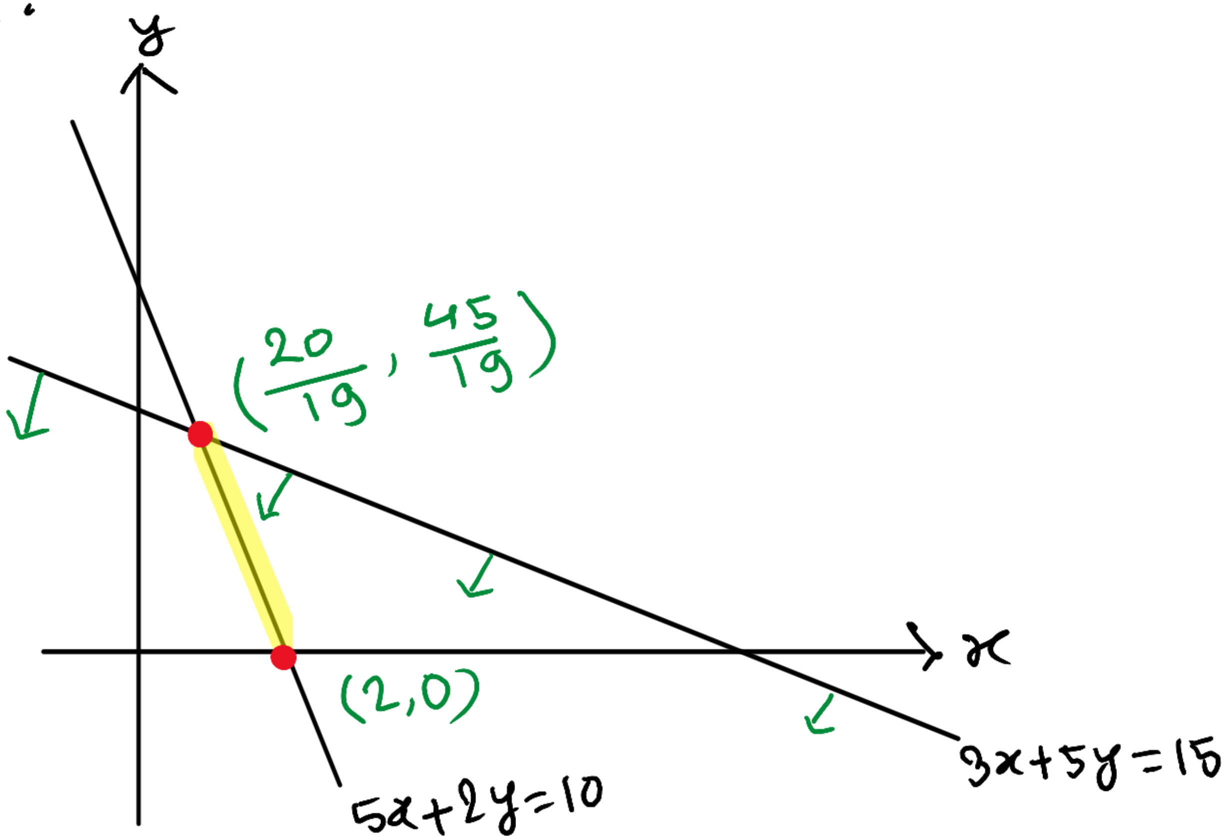
$$Z_{max} = 10.$$

Ex: Max $Z = 19x + 38y$

Sub to: $3x + 5y \leq 15$

$5x + 2y = 10$, $x \geq 0$, $y \geq 0$.

Sol^n:



$Z(2,0) = 38$

$Z\left(\frac{20}{19}, \frac{45}{19}\right) = 110$

The optimal sol^n is

$x = \frac{20}{19}$, $y = \frac{45}{19}$

$Z_{max} = 110$.